1. Create a bash script to **check if a directory is available or not**.

Step:1 :-

nano check_directory.sh

step:2 :-

#!/bin/bash

echo "Enter directory name:"

read dir

if [-d "$dir"]; then

  echo "Directory exists."

else

  echo "Directory does not exist."

Fi

 Press Ctrl + O and press Enter.

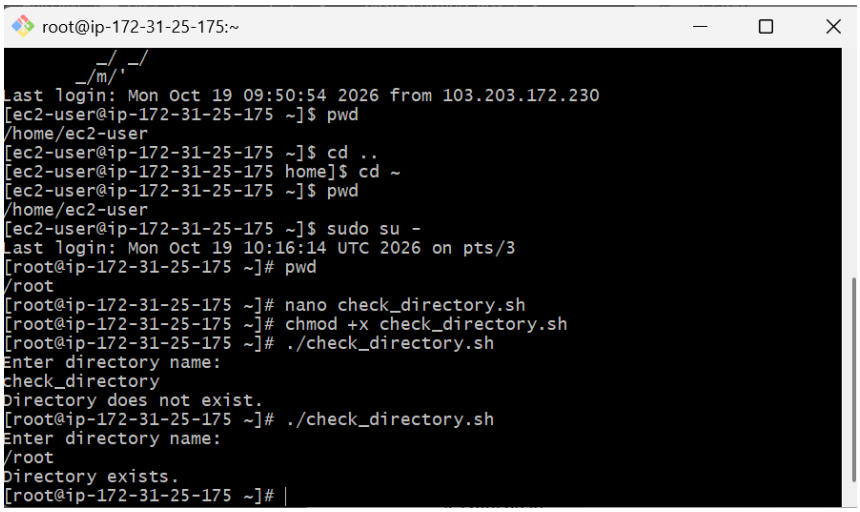
 Press Ctrl + X to exit.

Step:3

chmod +x check_directory.sh

step :4: Execute the script

./check_directory.sh



2. Create a bash script to **create multiple files**.

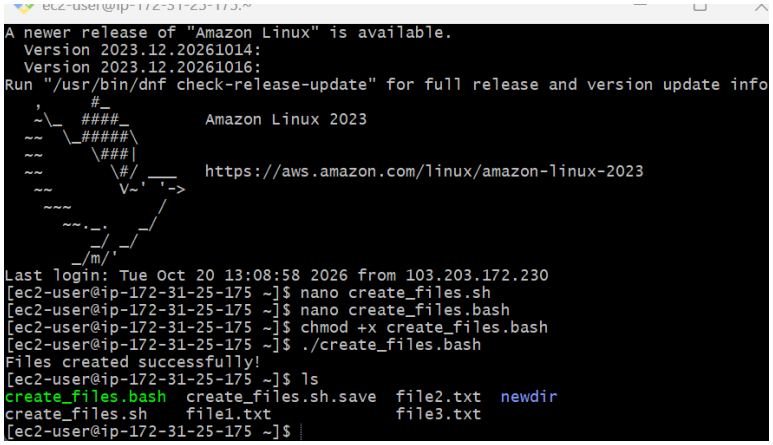
Step :1 : nano create_files.bash

Step:2 : #!/bin/bash

touch file1.txt file2.txt file3.txt

echo "Files created successfully!"

step:3 ./create_files.sh



3. Create a bash script to **take a backup of a directory**.

Step:1

mkdir mydata

Step:2

touch mydata/file1.txt mydata/file2.txt

step:3

nano backup.bash

step:4

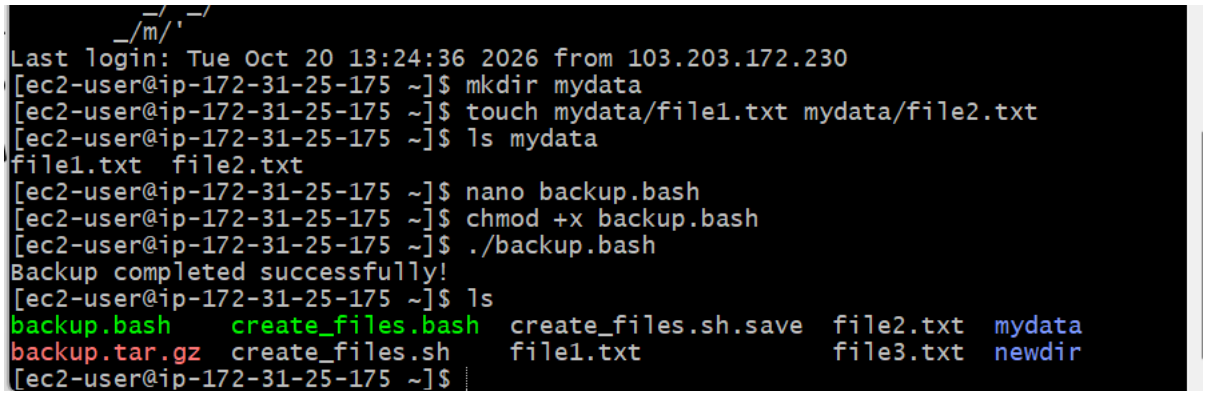
#!/bin/bash

tar -czf backup.tar.gz mydata

echo "Backup completed successfully!"

Step 5:

chmod +x backup.bash



3.

Step:1

nano install_nginx.bash

#!/bin/bash

sudo dnf install nginx -y

sudo systemctl start nginx

sudo systemctl enable nginx

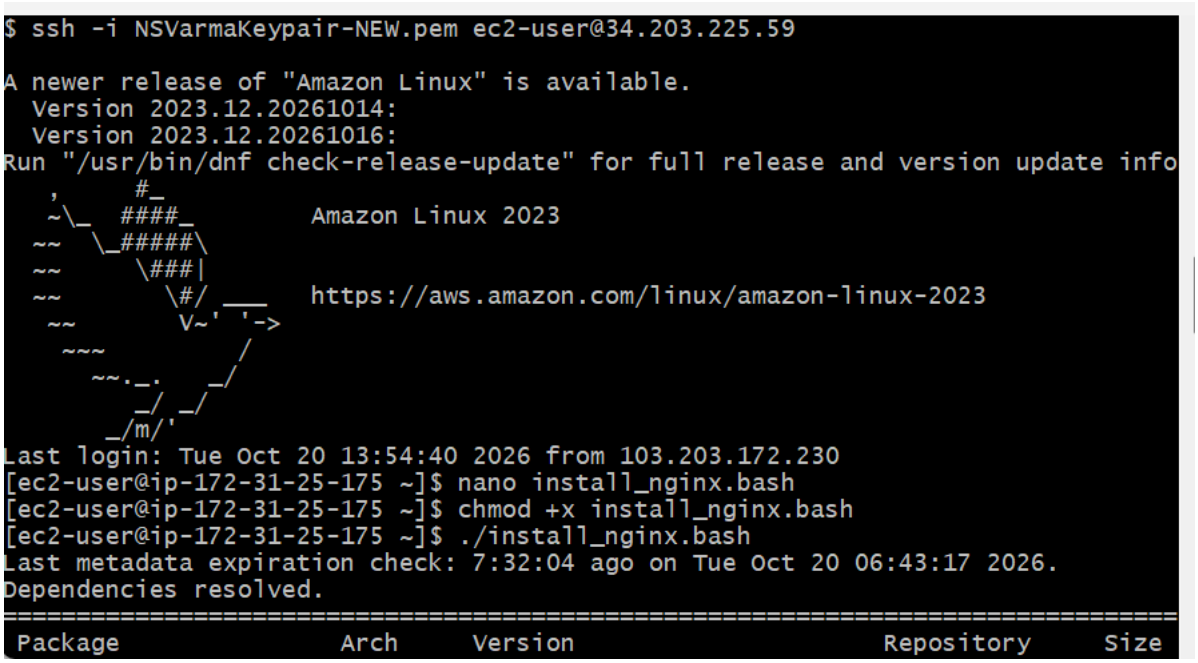
echo "Nginx installed successfully!"

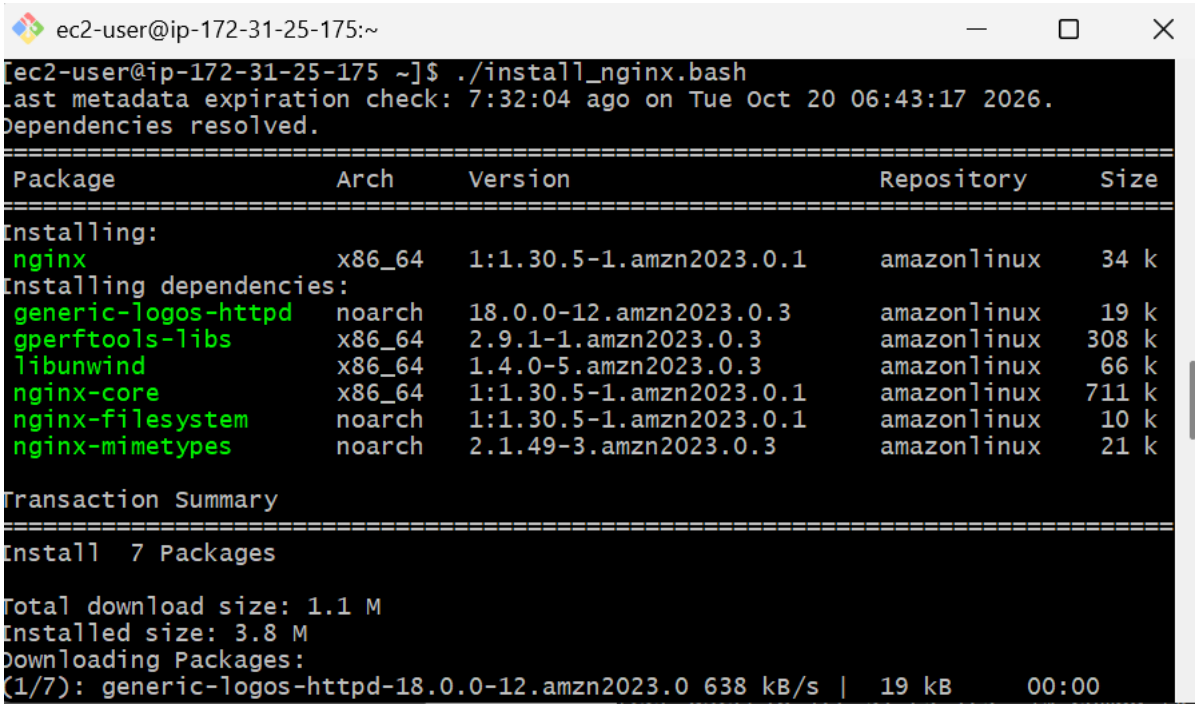
Step:2

chmod +x install_nginx.bash

Step:3

./install_nginx.bash





sudo systemctl status nginx

curl http://localhost

5.Create a bash script to **install Apache Tomcat on an EC2 server**.

nano install_tomcat.bash

#!/bin/bash

sudo dnf install java-17-amazon-corretto -y

cd /opt

sudo curl -O https://archive.apache.org/dist/tomcat/tomcat-10/v10.1.49/bin/apache-tomcat-10.1.49.tar.gz

sudo tar -xzf apache-tomcat-10.1.49.tar.gz

sudo mv apache-tomcat-10.1.49 tomcat

sudo /opt/tomcat/bin/catalina.sh start

echo "Tomcat installation completed!"

Step:

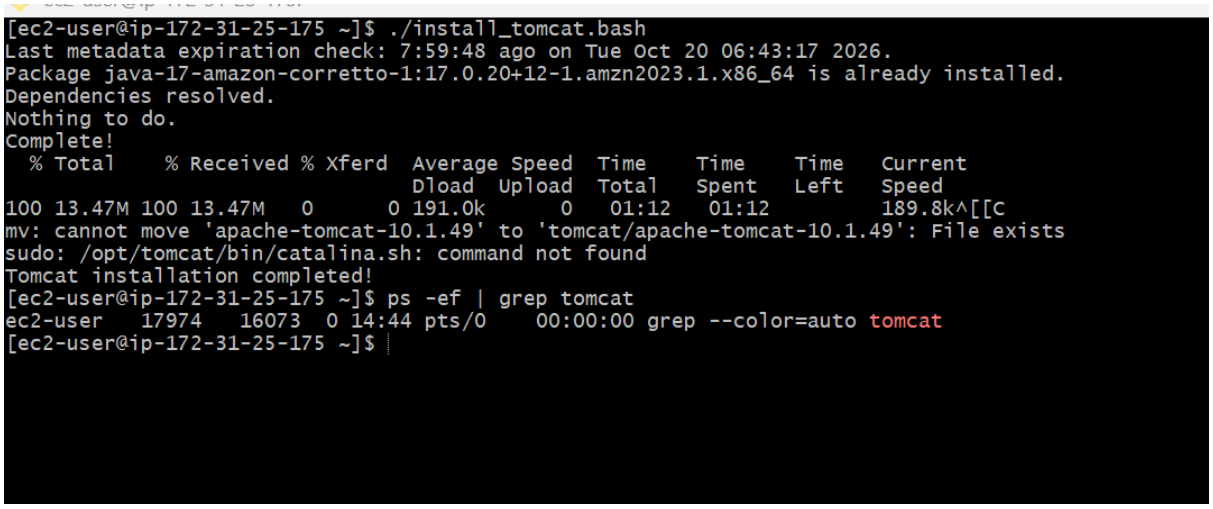
chmod +x install_tomcat.bash

Step:

./install_tomcat.bash

Step:

ps -ef | grep tomcat



6. Create a bash script to **check if the Nginx service is running**, if not running then script should start the service.

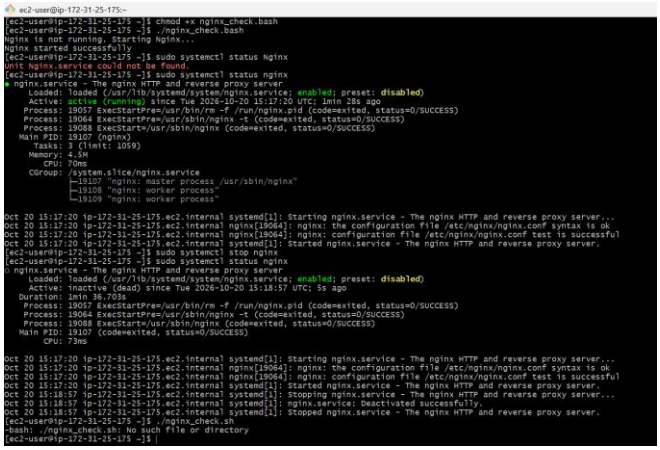
nano nginx_check.bash

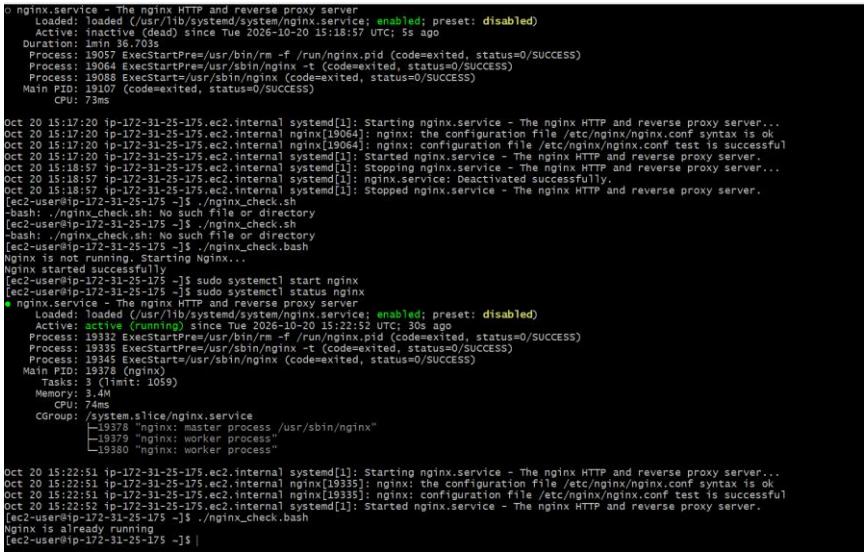
```
#!/bin/bash


if systemctl is-active --quiet nginx
then
   echo "Nginx is already running"
else
   echo "Nginx is not running. Starting Nginx..."
   sudo systemctl start nginx


   if systemctl is-active --quiet nginx
   then
      echo "Nginx started successfully"
   else
      echo "Failed to start Nginx"
   fi
fi
```

```
chmod +x nginx_check.bash
./nginx_check.bash
sudo systemctl status nginx
sudo systemctl start nginx
sudo systemctl stop nginx
```





7. Create a bash script for a **calculator**.

nano calculator.bash

#!/bin/bash

echo "Enter first number:"

read a

echo "Enter second number:"

read b

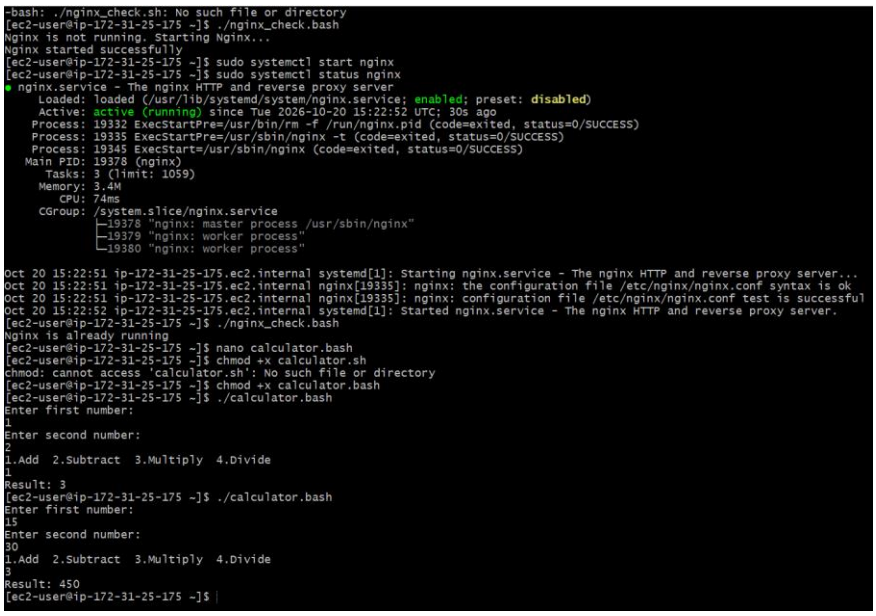
echo "1.Add 2.Subtract 3.Multiply 4.Divide"

```
read choice


if [ $choice -eq 1 ]; then
   echo "Result: $((a + b))"
elif [ $choice -eq 2 ]; then
   echo "Result: $((a - b))"
elif [ $choice -eq 3 ]; then
   echo "Result: $((a * b))"
elif [ $choice -eq 4 ]; then
   if [ $b -eq 0 ]; then
      echo "Cannot divide by zero"
   else
      echo "Result: $((a / b))"
   fi
else
   echo "Invalid choice"
fi
```

chmod +x calculator.sh

./calculator.bash



```
Oct 20 15:22:51 ip-172-31-25-175.ec2.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Oct 20 15:22:51 ip-172-31-25-175.ec2.internal nginx[19335]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Oct 20 15:22:51 ip-172-31-25-175.ec2.internal nginx[19335]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Oct 20 15:22:52 ip-172-31-25-175.ec2.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
[ec2-user@ip-172-31-25-175 ~]$ ./nginx_check.bash
Nginx is already running
[ec2-user@ip-172-31-25-175 ~]$ nano calculator.bash
[ec2-user@ip-172-31-25-175 ~]$ chmod +x calculator.sh
chmod: cannot access 'calculator.sh': No such file or directory
[ec2-user@ip-172-31-25-175 ~]$ chmod +x calculator.bash
[ec2-user@ip-172-31-25-175 ~]$ ./calculator.bash
Enter first number:
1
Enter second number:
2
1.Add  2.Subtract  3.Multiply  4.Divide
1
Result: 3
[ec2-user@ip-172-31-25-175 ~]$
```

8. Create a bash script to **check if a directory exists**, if not then create a directory.

nano directory.bash

```
#!/bin/bash

if [ -d "myfolder" ]; then
   echo "Directory already exists"
else
   mkdir myfolder
   echo "Directory created successfully"
fi
```

9. Create a bash script to **delete the last 3 lines of a file**.

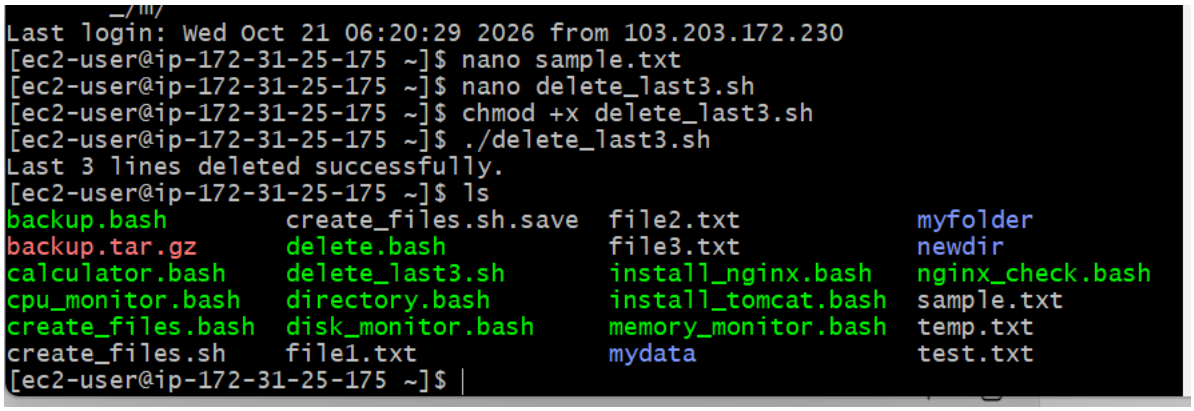
nano sample.txt

nano delete_last3.sh

chmod +x delete_last3.sh

./delete_last3.sh

Cat sample.txt



10.Bash script to monitor cpu and if it is more than 80% then send email notification.

nano cpu_monitor.sh

```
#!/bin/bash

THRESHOLD=80
EMAIL="your_email@example.com"

CPU_IDLE=$(mpstat 1 1 | awk '/Average/ && $NF ~ /^[0-9.]+$/ {print $NF}')

CPU_USAGE=$(awk -v idle="$CPU_IDLE" 'BEGIN {printf "%.0f", 100-idle}')
```

```
echo "Current CPU Usage: ${CPU_USAGE}%"

if [ "$CPU_USAGE" -gt "$THRESHOLD" ]; then
   echo "Warning: CPU usage is ${CPU_USAGE}%" |
      mail -s "High CPU Usage Alert" "$EMAIL"

   echo "Email notification sent."
else
   echo "CPU usage is normal."
Fi
chmod +x cpu_monitor.sh
./cpu_monitor.sh
```

11.Bash script to monitor disk space and if it is more than 80% then send email notification.

```
nano disk_monitor.sh

#!/bin/bash

THRESHOLD=80
EMAIL="your_email@example.com"

df -P -x tmpfs -x devtmpfs | awk 'NR > 1 {print $5, $6}' |
while read -r USAGE MOUNT
do
```

```
        USAGE=${USAGE%\%}

        if [ "$USAGE" -gt "$THRESHOLD" ]; then
          MESSAGE="Warning: Disk usage on $MOUNT is ${USAGE}%"

          echo "$MESSAGE"
          echo "$MESSAGE" | mail -s "High Disk Usage Alert" "$EMAIL"
        fi
      done
      chmod +x disk_monitor.sh
      ./disk_monitor.sh
```

12.Bash script to monitor memory and if it is more than 80% then send email notification.

```
nano memory_monitor.sh

#!/bin/bash

THRESHOLD=80
EMAIL="your_email@example.com"

MEMORY_USAGE=$(free | awk '/Mem:/ {
  printf "%.0f", ($2 - $7) / $2 * 100
}')

echo "Current Memory Usage: ${MEMORY_USAGE}%"

if [ "$MEMORY_USAGE" -gt "$THRESHOLD" ]; then
  MESSAGE="Warning: Memory usage is ${MEMORY_USAGE}%"

  echo "$MESSAGE" | mail -s "High Memory Usage Alert" "$EMAIL"
```

```
    echo "Email notification sent."
else
    echo "Memory usage is normal."
Fi
chmod +x memory_monitor.sh
./memory_monitor.sh
```

Exercise: Crontab Entries

1. April 5th Midnight

0 0 5 4 * /root/cronjob/april5.bash

2. 5th of Every November, January, June if it is a Thursday

0 0 5 1,6,11 4 /root/cronjob/thursday.bash

At first, I thought my above answer is correct.

0 0 5 1,6,11 * ["$(date +\%u)" -eq 4] && /root/cronjob/thursday.bash

3. At 05 and 27th minutes of 9, 10, 11 hours every day

5,27 9,10,11 * * * /root/cronjob/everydayjob.bash

4. 34 minutes of 9th hour on 15th August

34 9 15 8 * /root/cronjob/15august.bash

5. Every midnight

0 0 * * * /root/cronjob/everymidnyt.bash

6. Every Weekend (Saturday night 11:59 PM)

59 23 * * 6 /root/cronjob/weeklysatjob.bash

7. After every reboot

@reboot /root/cronjob/rebootjob.bash